

ZURICH AP, Switzerland

WMO: 066700

Lat: 47.4797N

Lon: 8.5358E

Elev: 432

StdP: 96.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.5	(d) -12.1	(e) 1.4	(f) -7.1	(g) -9.6	(h) 1.7	(i) -5.2	(j) 10.9	(k) 8.3	(l) 9.7	(m) 7.2	(n) 2.2	(o) 40	(p) 0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	30.9	20.1	28.9	19.3	27.0	18.7	20.9	28.4	20.0	27.0	19.2	25.5	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.0	23.5	17.7	13.4	22.5	17.0	12.8	21.7	62.2	28.6	59.2	27.2	56.5	25.6	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 6.9	(c) 5.8	(e) -12.3	(f) 33.4	(g) 2.9	(h) 1.7	(i) -14.4	(j) 34.6	(k) -16.1	(l) 35.6	(m) -17.7	(n) 36.6	(o) -19.8	(p) 37.8		
(a) 8.2	(b) 6.9	(c) 5.8	(e) -12.7	(f) 22.7	(g) 3.0	(h) 0.7	(i) -14.9	(j) 23.2	(k) -16.6	(l) 23.6	(m) -18.3	(n) 24.0	(o) -20.5	(p) 24.5		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	0.9	1.9	5.9	9.7	13.9	17.6	19.4	18.9	14.6	10.4	4.9	1.7
(7)		DBStd	7.55	4.07	4.21	3.84	3.71	3.39	3.61	3.12	3.11	3.17	3.59	3.65	3.88
(8)		HDD10.0	1161	283	228	135	51	7	1	0	0	3	39	156	259
(9)		HDD18.3	3183	541	462	387	259	141	56	25	30	117	245	403	517
(10)		CDD10.0	1172	1	0	7	41	128	229	292	277	142	52	3	0
(11)		CDD18.3	153	0	0	0	0	4	35	58	49	6	0	0	0
(12)		CDH23.3	1566	0	0	0	9	84	377	574	463	57	1	0	0
(13)		CDH26.7	463	0	0	0	1	11	109	186	152	4	0	0	0
(14)	Wind	WSAvg	2.2	2.5	2.6	2.7	2.4	2.3	2.1	2.1	1.8	2.0	2.0	2.1	2.4
(15)	Precipitation	PrecAvg	1207	69	72	75	88	131	136	140	132	97	88	90	89
(16)		PrecMax	1541	170	162	227	175	243	222	257	236	208	159	195	185
(17)		PrecMin	950	9	21	24	12	30	51	53	18	45	8	0	1
(18)		PrecStd	154	42	38	44	43	58	44	54	52	40	40	49	47
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	12.2	14.6	20.0	24.7	28.1	32.3	33.4	33.7	27.1	22.0	16.3	12.1	
(20)		0.4% MCWB	8.9	8.5	11.5	13.9	17.5	20.7	20.3	20.2	18.7	15.7	11.7	9.1	
(21)		2% DB	10.5	12.0	17.2	22.1	25.8	29.7	30.9	30.4	25.1	19.8	13.6	10.3	
(22)		2% MCWB	7.7	7.4	10.0	12.9	16.7	19.6	20.0	20.0	17.7	14.9	10.5	8.1	
(23)		5% DB	8.4	10.1	14.9	19.9	23.5	27.6	28.7	28.1	22.8	17.7	11.6	8.9	
(24)		5% MCWB	6.2	6.7	9.4	12.2	15.3	18.9	19.3	19.4	16.7	14.0	9.4	7.0	
(25)		10% DB	6.8	8.1	12.7	17.5	21.3	25.3	26.6	25.9	20.8	15.9	10.1	7.3	
(26)		10% MCWB	5.1	5.5	8.2	11.0	14.5	17.9	18.6	18.6	16.0	13.1	8.4	5.8	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	9.7	9.4	12.2	14.8	18.8	22.0	22.0	21.9	19.7	16.7	12.3	9.9	
(28)		0.4% MCDB	11.9	12.9	18.0	22.7	26.3	29.9	30.6	29.7	25.4	20.6	15.1	11.4	
(29)		2% WB	7.9	8.0	10.8	13.5	17.4	20.5	20.7	20.8	18.4	15.5	10.9	8.4	
(30)		2% MCDB	10.0	11.5	15.4	20.5	24.0	27.6	28.4	28.1	23.6	18.7	13.0	10.0	
(31)		5% WB	6.4	6.9	9.7	12.4	16.2	19.5	19.9	19.9	17.4	14.5	9.7	7.2	
(32)		5% MCDB	8.2	9.7	13.9	18.7	21.9	26.0	27.0	26.5	21.8	17.1	11.4	8.6	
(33)		10% WB	5.1	5.7	8.6	11.4	15.1	18.4	19.0	19.1	16.4	13.4	8.5	5.8	
(34)		10% MCDB	6.6	8.0	12.3	16.8	20.0	24.3	25.2	25.0	20.0	15.5	9.9	7.1	
(35)	Mean Daily Temperature Range	MDBR	5.4	7.6	9.8	11.0	10.7	11.0	11.1	10.8	9.9	8.2	6.1	5.1	
(36)		5% DB MCDBR	7.9	11.8	14.6	16.4	15.6	15.2	15.4	15.3	13.2	11.1	8.7	7.2	
(37)		5% WB MCWBR	5.8	7.7	8.4	8.3	7.4	6.4	6.0	6.2	6.7	6.7	6.0	5.3	
(38)		5% WB MCDBR	7.3	10.4	12.7	14.6	13.7	13.9	13.8	13.4	11.9	10.0	8.1	6.5	
(39)		5% WB MCWBR	5.5	7.1	7.5	7.5	6.7	6.1	5.7	5.9	6.3	6.2	5.8	5.1	
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.322	0.358	0.392	0.405	0.409	0.411	0.404	0.378	0.353	0.320	0.293	
(41)		taud	2.452	2.379	2.332	2.267	2.271	2.300	2.289	2.325	2.377	2.433	2.475	2.457	
(42)		Ebn at Noon	792	839	858	860	860	856	848	839	827	793	754	750	
(43)		Edh at Noon	66	87	106	125	129	126	126	117	101	82	64	58	
(44)	All-Sky Solar Radiation	RadAvg	1.14	1.98	3.19	4.33	5.00	5.73	5.53	4.83	3.62	2.27	1.26	0.94	
(45)		RadStd	0.12	0.27	0.43	0.62	0.50	0.53	0.51	0.50	0.37	0.24	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A	N/A
(47) Regional (116 neighbors)	N/A	N/A	N/A	+0.70	+0.42	+0.34	N/A	N/A	+56	+15

Nomenclature: See separate page